Animated Plots

1. A Motivating Example
   - drawing the golden rectangle
2. Animating Two Dimensional Plots
   - increasing the frequency of a periodic function
   - a moving tangent line to the unit circle
3. Animating Three Dimensional Plots
   - spinning a surface
   - a space curve knot

MCS 320 Lecture 28
Introduction to Symbolic Computation
Jan Verschelde, 20 July 2026

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The Golden Rectangle

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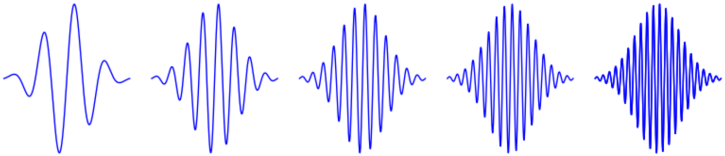
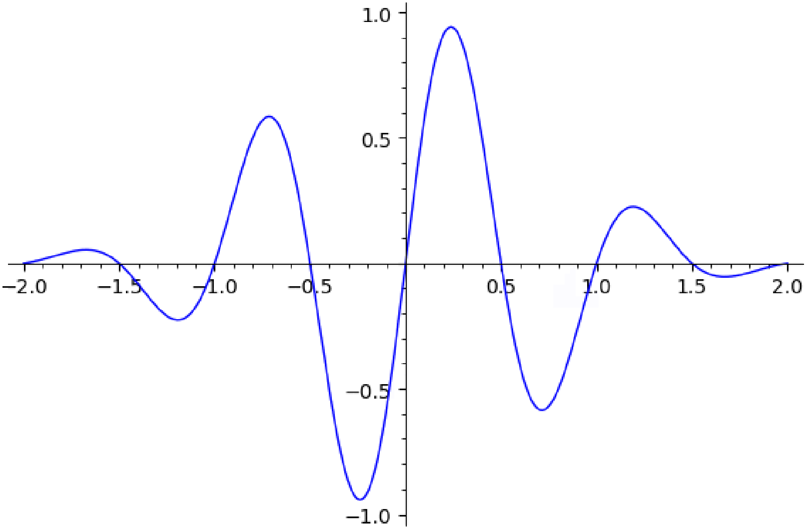
Increasing the Frequency of a Periodic Function

Consider $y = \exp(-x^2)\sin(2\pi x)$.

- Introduce a parameter k for the frequency in $\exp(-x^2)\sin(2k\pi x)$.
- The first five frames are plots for $k = 1, 2, 3, 4, 5$.



Animating the Frequency



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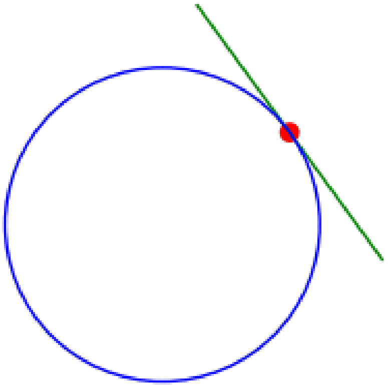
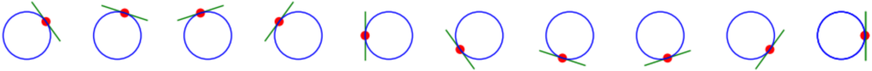
Animating the Tangent Line to the Unit Circle



The Components in the Animation



Three components in this animation:

1. The unit circle remains static.
2. The moving point is shown as a red disk.
3. The moving tangent is the green line.

While each frame is computed separately, they all share

1. the same static unit circle, and
2. the same layout for the point and line.

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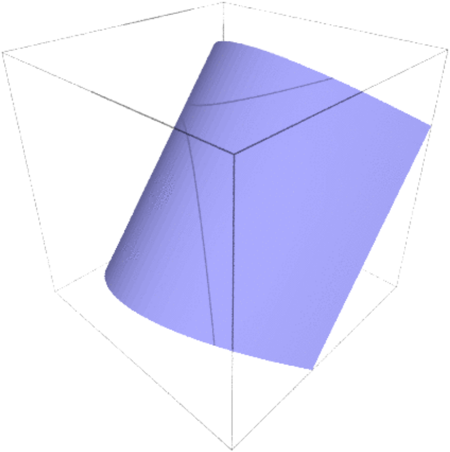
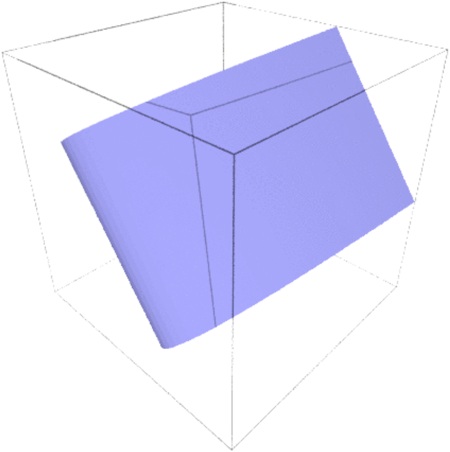
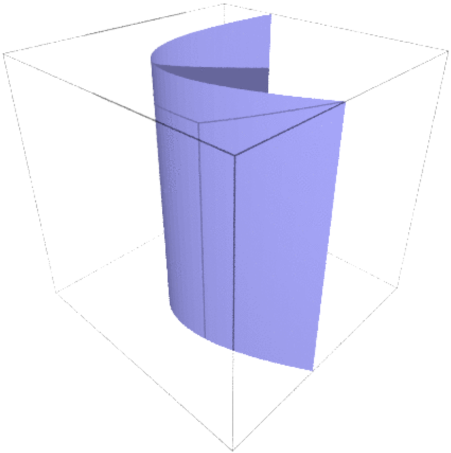
Spin $x^2 - y = 0$ by Rotating the x-Axis



Spin $x^2 - y = 0$ by Rotating the y-Axis



Spin $x^2 - y = 0$ by Rotating the z-Axis



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A Space Curve Knot